

Samuel Babichenko

sam@sbabichenko.com · sbabichenko.com · github.com/sbabichenko · Charlotte, NC

Education

PhD in Statistics and Applied Probability, UC Santa Barbara – Sep 2021 – Jun 2026

Advisor: Tomoyuki Ichiba. Thesis: *Noise-State Calculus for Dynamic Games with Strategic Information*.

Passed all qualifying exams prior to the first term of the PhD.

Graduate Teaching Assistant. Mentored 10+ undergraduates, majority of whom are now in PhD programs. Presented at UCSD Stochastic Systems Seminar.

BS in Mathematics, UC San Diego – Sep 2019 – Jun 2021

Dean's Undergraduate Excellence Award (top 0.5%, Physical Sciences).

Research

Forecasting and Manipulating the Forecasts of Others – arXiv: 2603.12140

Solo-authored. Submitted March 2026.

First exact solution to a class of multi-agent games with private signals open since Townsend (1983). Solves for equilibrium strategies, price impact, and welfare costs in markets where agents learn from each other's actions. Works in the exact model class behind Kyle-type market microstructure, optimal execution, and multi-agent control, with no approximation or large-population limit.

Quantifies how information asymmetry distorts prices and behavior: not just added noise, but systematic bias in mean prices, excess volatility, and wasted effort from strategic belief manipulation. Decomposes the total cost into an estimation component and a strategic component, showing the strategic channel dominates by an order of magnitude. Identifies welfare reversals where more precise private information raises total cost—a non-monotone equilibrium effect absent from cooperative or single-agent models.

C++ solver computes equilibria in ~10ms. Compiled to WebAssembly for a browser-side interactive explorer at ~60ms. Live demo: sbabichenko.com/noisestate. Python package: `pip install noisestate`.

Experience

Wells Fargo, Charlotte, NC – *Quantitative Researcher* – Jul 2026 – present

Wells Fargo, Charlotte, NC – *Quantitative Researcher Intern, Mortgage Model Development* – Jun – Aug 2025

- Completed assigned mortgage pool segmentation project in two days; independently built an improved tree-based alternative with diagnostic tools in one week. Work led the buy-side trading desk to pull me onto a harder open-ended problem.
- With no prior MBS background, identified a structural flaw in the team's prepayment models: normal approximations miss the binomial structure of discrete loan-level prepayment. Built a binomial model correcting for effective loan count, partitioned pools to isolate systemic errors, and found them highly persistent ($R^2 \approx 0.8$ at 5-month horizon), converting model misspecification into a tradeable signal for a \$200B portfolio.

Selected Projects

Multi-Agent Traffic Simulation (Waymo Open Motion Dataset)

Reimplemented BehaviorGPT for multi-agent traffic simulation: decoder-only transformer with agent-agent attention, relative spacetime embeddings, and next-patch prediction generating closed-loop 10 Hz trajectories on the Waymo Open Motion Dataset.

Extending to a two-stage architecture: the transformer predicts only at branch points where agent behavior becomes multimodal (lane changes, yielding decisions, intersection entries), then a conditional diffusion model interpolates realistic trajectories between branch points with guided sampling.

Adaptive Triangle-Mesh Regression

Continuous surface model on an adaptive right-triangle bisection mesh with hierarchical partial pooling and wavelet shrinkage, addressing the discontinuity problem of tree-based models observed in the Wells Fargo prepayment project. Runs in the browser: sbabichenko.com/gate.

Skills

Programming: C++23, Python, CUDA, Triton, Linux, bash, SQL, computer architecture

ML & Data: Transformer architectures, diffusion models, neural networks for control, PyTorch, NumPy, Polars, Pandas, LightGBM, XGBoost

Methods: Stochastic calculus, stochastic control, Kalman filtering, Monte Carlo, PDE methods, time series, derivatives pricing, Bayesian inference, causal inference, experimental design